

Solution of Power BI Assignment – DAX for New Insights

1. Load Data into Power BI (5 Points)

Steps:

1. Open Power BI Desktop.
2. Click on Home -> Get Data -> Text/CSV.
3. Browse to the CSV file containing the Product Sales Data and click Open.
4. Ensure that the data is loaded correctly by checking that all columns appear properly in the Power BI data model.

Expected Output:

- The data is displayed correctly in the Power BI Data pane.
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2. Create a DAX Measure for Total Sales (10 Points)

Steps:

1. In the Power BI Fields pane, right-click the table with your data (e.g., Product Sales Data) and select New Measure.
2. Create a DAX measure for Total Sales. The formula should be:

```
Total Sales = SUM('Product Sales Data'[Sales])
```

3. Drag the Total Sales measure into a Card visualization to display the total sales.

Expected Output:

- A Card visualization showing the total sales.
-

3. Create a DAX Measure for Average Sales Per Product (15 Points)

Steps:

1. Right-click the table and select New Measure to create a new DAX measure.

```
Average Sales Per Product =  
DIVIDE(SUM('Product Sales Data'[Sales]), DISTINCTCOUNT('Product Sales  
Data'[Product ID]))
```

2. Drag the Average Sales Per Product measure into another Card visualization to display the result.

Expected Output:

- A Card visualization showing the average sales per product.
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4. Create a DAX Measure for Yearly Sales Growth Percentage (15 Points)

Steps:

1. Right-click the table and select `New Measure` to create a new DAX measure.
2. Use the following DAX formula to calculate the Yearly Sales Growth Percentage:

```
Yearly Sales Growth % =  
VAR CurrentYearSales = SUM('Product Sales Data'[Sales])  
VAR PreviousYearSales =  
    CALCULATE(  
        SUM('Product Sales Data'[Sales]),  
        SAMEPERIODLASTYEAR('Product Sales Data'[Order Date])  
    )  
RETURN  
DIVIDE(CurrentYearSales - PreviousYearSales, PreviousYearSales)
```

3. Create a Bar Chart and place `Order Date` on the Axis and `Yearly Sales Growth %` on the Value.

Expected Output:

- A Bar Chart visualizing Yearly Sales Growth Percentage.
-

5. Create a DAX Measure for Total Profit (15 Points)

Steps:

1. Right-click the table and select `New Measure` to create a new DAX measure.
2. Use the following DAX formula to calculate Total Profit:

```
Total Profit = SUM('Product Sales Data'[Profit])
```

3. Create a Table visualization with `Product Name`, `Category`, and `Total Profit` displayed.

Expected Output:

- A Table visualization displaying the Total Profit alongside Product Name and Category.
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6. Create a DAX Measure for Profit Margin (15 Points)

Steps:

1. Right-click the table and select `New Column` to create a new DAX calculated column.
2. Use the following DAX formula to calculate Profit Margin:

```
Profit Margin = DIVIDE('Product Sales Data'[Profit], 'Product Sales Data'[Sales])
```

3. Create a Table visualization with `Product Name`, `Sales`, and `Profit Margin`.

Expected Output:

- A Table visualization displaying the Profit Margin alongside Product Name and Sales.
-

7. Create Visualizations to Display Insights (20 Points)**Steps:**

1. **Total Sales by Region:**
 - Create a Bar Chart visualization.
 - Place `Region` on the Axis and `Total Sales` on the Values.
 - Add a title like "Total Sales by Region."
2. **Total Profit by Category:**
 - Create a Pie Chart or another suitable chart.
 - Place `Category` on the Legend and `Total Profit` on the Values.
 - Add a title like "Total Profit by Category."

Expected Output:

- A Bar Chart showing Total Sales by Region.
 - A Pie Chart (or another chart type) showing Total Profit by Category.
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8. Publish and Share the Report (5 Points)**Steps:**

1. Save your Power BI report as a `.PBIX` file.
2. Click `File -> Publish` to publish the report to Power BI Service.
3. Share the report link with your instructor or submit the `.PBIX` file directly.

Expected Output:

- The report is shared with the instructor.
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Summary of DAX Formulas:

1. Total Sales:

```
Total Sales = SUM('Product Sales Data'[Sales])
```

2. Average Sales Per Product:

```
Average Sales Per Product =  
DIVIDE(SUM('Product Sales Data'[Sales]), DISTINCTCOUNT('Product Sales  
Data'[Product ID]))
```

3. Yearly Sales Growth %:

```
Yearly Sales Growth % =  
VAR CurrentYearSales = SUM('Product Sales Data'[Sales])  
VAR PreviousYearSales =  
    CALCULATE(  
        SUM('Product Sales Data'[Sales]),  
        SAMEPERIODLASTYEAR('Product Sales Data'[Order Date])  
    )  
RETURN  
DIVIDE(CurrentYearSales - PreviousYearSales, PreviousYearSales)  
Total Profit = SUM('Product Sales Data'[Profit])
```

4. Profit Margin (Calculated Column):

```
Profit Margin = DIVIDE('Product Sales Data'[Profit], 'Product Sales  
Data'[Sales])
```